Lecture - 5

CO4: Compare various statistical methods of study data samples

CO5: Analyze and evaluation of different sets of data using hypothesis testing.

BAYES' THEOREM

We have discussed that if we are interested in finding the probability of the event of second stage, then it is obtained using law of total probability. But if the happening of the event of second stage is given to us and on this basis, we find the probability of the events of first stage, then the probability of an event of first stage is the revised (or posterior) probabilities and is obtained using an important theorem known as Bayes' theorem given by Thomas Bayes (died in 1761, at the age of 59), a British Mathematician, published after his death in 1763. This theorem is also known as 'Inverse probability theorem', because here moving from first stage to second stage, we again find the probabilities (revised) of the events of first stage i.e., we move inversely. Thus, using this theorem, probabilities can be revised on the basis of having some related new information. As an illustration, let us consider the same example as taken in Sec. 4.2 of this unit. In this example, if we are given that the drawn ball is of particular colour and it is asked to find, on this basis, the probability that Urn I or Urn II was selected, then these are the revised (posterior) probabilities and are obtained using Bayes' theorem, which is stated and proved as under:

Statement: Let S be the sample space and E_1, E_2, …, E_n be n mutually exclusive and exhaustive events with $P(E_i) \neq 0$; i = 1, 2, ..., n. Let A be any event which is a sub-set of $E_1 \cup E_2 \cup \ldots \cup E_n$ (i.e. at least one of the events E_1, E_2, …, E_n) with P(A) > 0 [Notice that up to this line the statement is same as that of law of total probability], then

$$P(E_i|A) = \frac{P(E_i)P(A|E_i)}{P(A)}, i = 1,2, \ldots, n$$

Where $P(A) = P(E_1)\,P(A|E_1) + P(E_2)\,P(A|E_2) + \ldots + P(E_n)\,P(A|E_n)$.

Proof: First you have to prove that

$P(A) = P(E_1)\,P(A|E_1) + P(E_2)\,P(A|E_2) + \ldots + P(E_n)\,P(A|E_n)$.

Which is the law of total probability, After this

$$P(E_i|A) = \frac{P(E_i \cap A)}{P(A)},$$

[Applying the formula of conditional probability i.e., $P(A|B) = \frac{P(A \cap B)}{P(B)}$]

$$P(E_i|A) = \frac{P(E_i)P(A|E_i)}{P(A)},$$

[Applying multiplication theorem for dependent event i.e., $P(B|A)P(A) = P(A \cap B)$]

APPLICATIONS OF BAYES' THEOREM

Example 1: There are two bags. First bag contains 5 red, 6 white balls and the second bag contains 3 red, 4 white balls. One bag is selected at random and a ball is drawn from it.

If it is found to be red, what is the probability of?

i) selecting the first bag

ii) selecting the second bag

Solution: Let E_1 be the event that first bag is selected and E_2 be the event that second bag is selected.

$\therefore$ $P(E_1) = P(E_2) = 1/2$.

i) Probability of selecting the first bag given that the ball drawn is red

$$= P(E_1|R)$$

$$= \frac{P(E_1)P(R|E_1)}{P(R)},$$

$$= \frac{\frac{1}{2} * \frac{5}{11}}{\frac{34}{77}} = \frac{5}{22} * \frac{77}{34} = \frac{35}{68}$$

i) Probability of selecting the second bag given that the ball drawn is red

$$= P(E_2|R)$$

$$= \frac{P(E_2)P(R|E_2)}{P(R)},$$

$$= \frac{\frac{1}{2} * \frac{3}{7}}{\frac{34}{77}} = \frac{3}{14} * \frac{77}{34} = \frac{33}{68}.$$

Example 2: A factory produces certain type of output by 3 machines. The respective daily production figures are-machine X: 3000 units, machine Y: 2500 units and machine Z: 4500 units. Past experience shows that 1% of the output produced by machine X is defective. The corresponding fractions of defectives for the other two machines are 1.2 and 2 percent respectively. An item is drawn from the day's production. If the drawn item is found to be defective, what is the probability that it has been produced by machine Y?

Solution: Let E_1, E_2 and E_3 be the events that the drawn item is produced by machine X, machine Y and machine Z, respectively. Let A be the event that the drawn item is defective.

As the total number of units produced by all the machines is

3000 + 2500 + 4500 = 10000,

$$\therefore P(E_1) = \frac{3000}{10000} = \frac{3}{10}, P(E_2) = \frac{2500}{10000} = \frac{1}{4}, P(E_3) = \frac{4500}{10000} = \frac{9}{20}.$$

$$P(A|E_1) = \frac{1}{100} = 0.01, P(A|E_2) = \frac{1.2}{100} = 0.012, P(A|E_3) = \frac{2}{100} = 0.02.$$

Probability that the drawn item has been produced by machine Y given that it is defective

$$= P(E_2|A)$$

$$= \frac{P(E_2)P(A|E_2)}{P(A)},$$

$$= \frac{\frac{1}{4} * 0.012}{0.015} = \frac{0.003}{0.015} = \frac{1}{5}.$$

Now, you can try the following exercises.

1. A bag contains 4 red and 5 white balls. Another bag contains 2 red and 3 white balls. A ball is drawn from the first bag and is transferred to the second bag. A ball is then drawn from the second bag and is found to be red, what is the probability that red ball was transferred from first to second bag?
2. An insurance company insured 1000 scooter drivers, 3000 car drivers and 6000 truck drivers. The probabilities that scooter, car and truck drivers meet an accident are 0.02, 0.04, 0.25 respectively. One of the insured persons meets with an accident. What is the probability that he is a

   (i) car driver (ii) truck driver.
3. By examining the chest X-ray, the probability that T.B is detected when a person is actually suffering from T.B. is 0.99. The probability that the doctor diagnoses incorrectly that a person has T.B. on the basis of X-ray is 0.002. In a certain city, one in 1000 persons suffers from T.B. A person is selected at random and is diagnosed to have T.B., what is the chance that he actually has T.B.?
4. A person speaks truth 3 out of 4 times. A die is thrown. She reports that there is five. What is the chance there was five?